3rd Semester

Data Science Foundations Lab Syllabus

1. Write a program in python to print “Hello world” message on the screen.
2. Write a program in python to print student Bio data on the screen.
3. Write a program in python to add two number and display their result on the screen.
4. Write a program in python to swap two numbers using a third variable.
5. Write a program in python to find the square root of a number.
6. Write a program in python to calculate the area of a triangle.
7. Write a program in python to solve a quadratic equation.
8. Write a program in python to perform command line argument.
9. Write a program in python to check whether the number is +ve or -ve using if-else statement.
10. Write a program in python to check whether the number is +ve or -ve using if-elif-else statement.
11. Write a program in python to calculate the summation of 10 number using while loop statement.
12. Write a program in python to print 1 to 10 using for loop statement.
13. Write a program in python to add number from 1 to 20.
14. Write a program in python to print all even number between 1 to 20.
15. Write a program in python to check whether a number is equals with another number by using break statement.
16. Write a program in python to print the even number from 1 to 10 by using continue statement.
17. Write a program in python to check whether a particular number is equal with another number by using the pass statement.
18. Write a program in python to take a string after that you access at least one sub-string from the given string.
19. Write a program in python to perform multiline string literals.
20. Write a program in python to display a sub string from original string.
21. Write a program in python to perform the mathematical operation over string.
22. Write a program in python to replace a new string in place of the original string.
23. Write a program in python to split a string.
24. Write a program in python to perform join operation like -,*,/,^.
25. Write a program in python to perform following string conversion i) upper ii) lower iii) swap iv) title v) capitalize
26. Write a program in python to check starting word of a string from the original string.
27. Write a program in python to check end of the string from the original string.
28. Write a program in python to do following – “My name is BCET, I born in 2001”
29. Write a program in python to format a string using % symbol.
30. Write a program in python to format a string using template class.
31. Write a program in python to create a list of 5 element and access each element by using their index number.
32. Write a program in python to perform list operation like append( ), extend( ), insert( ), remove( ), pop( ) and len( ).
33. Write a program in python to take a list from user and sort in reverse order.
34. Write a program in python to take heterogenous or unordered list and sort it to ascending order.
35. Write a program in python to take a list string type and sort it to dictionary order.
36. Write a program in python to take a set and print the class.
37. Write a program in python to take a set {7,8,9.0.7} and add at least three element in it.
38. Write a program in python to take two set and perform update operation.
39. Write a program in python to take a set and perform copy operation.

40. Write a program in python to take a set form pop and clear operation.
41. Write a program in python to perform different mathematical set operation like union, intersection, difference and symmetric difference.
42. Write a program in python to take a string in the form of set and check particular character string is present or not.
43. Write a program in python to create a tuple and check the class name
44. Write a program in python to take a list and print particular element of the tuple.
45. Write a program in python to display set of elements using slice of operator.
46. Write a program in python to demonstrate mathematical operator like concatenation operator.
47. Write a program in python to perform following mathematical function in a tuple like lens( ), count( ), index( ), sorted( ) , min( ) and max( ).
48. Write a program in python to take a dictionary and print the content of that dictionary.
49. Write a program in python to add an item to the existing dictionary.
50. Write a program in python to remove an item from the dictionary.
51. Write a program in python to take a dictionary and update item value and print the updated dictionary.
52. Write a program in python to display “Hello” message using function.
53. Write a program in python to calculate square of a particular number taking input from the user by using function.
54. Write a program in python to check whether a number is even or odd using function.
55. Write a program in python to add two number using function.
56. Write a program in python to demonstrate keyword as argument using function.
57. Write a program in python to demonstrate default argument using function.
58. Write a program in python to demonstrate variable length argument using function.
59. Write a program in python to calculate factorial of a number using function.
60. Write a program in python to add two number using lambda function.
61. Write a program in python to print square of a number using lambda function.
62. Write a program in python to remove a particular character from string.
63. Write a program in python to declare a number list and find only odd number using lambda function.
64. Write a program in python take number list and square each element in the list.
65. Write a program in python to take a list by using reduce( ) function to calculate the summation of that list.
66. Write a program in python to take a list by using reduce( ) to calculate the division of that list with each element.
67. Write a program in python to check the function identity.
68. Write a program in python to perform following operation using nested function 1) addition 2) multiplication 3) Square root.
69. Write a program in python to add two number using decorator function.
70. Write a program in python to add all even and odd number from 1 to 100.
71. Write a program in python to demonstrate member aliasing and module aliasing.
72. Write a program in python to perform addition operation over two array in Numpy.
73. Write a program in python to implement array operation using function in Numpy.
74. Write a program in python to take an array and find the mean and median of that array.
75. Write a program in python to take two array and perform dot( ) product of two array.
76. Write a program in python to take an array and access particular element in array using numpy.
77. Write a program in python to take an array and convert it to series, after that print the result.
78. Write a program in python to take to create custom index using pandas series.
79. Write a program in python to create a DataFrame from the dictionary.

80. Write a program in Python to find a list of all names in the following string “Amy is 5 years old, and her sister Mary is 2 years old. Ruth and Peter, their parents, have 3 kids.” using regex.
81. The dataset file in assets/grades.txt contains a line-separated list of people with their grades in a class. Write a program in Python to create a regex to generate a list of just those students who received a B in the course.
82. Consider the standard web log file in assets/logdata.txt. This file records the access a user makes when visiting a web page. Each line of the log has the following items: • a host • a user_name • the time a request was made • the post request type
    Our task is to convert this into a list of dictionaries using Python Programming Language, where each dictionary looks like the following: example_
    dict = {"host":"146.204.224.152", "user_name":"feest6811", "time":"21/Jun/2019:15:45:24 -0700", "request":"POST /incentivize HTTP/1.1"} Write a program in python to implement all trigonometric function using Numpy.
83. Download the csv data file from(assets/NISPUF17.csv) and Write a function called proportion_of_education which returns the proportion of children in the dataset who had a mother with the education levels equal to less than high school (12) and college degree.
84. Download the csv data file from(assets/NISPUF17.csv) and write a program in python to explore the relationship between being fed breastmilk as a child and getting a seasonal influenza vaccine from a healthcare provider. Return a tuple of the average number of influenza vaccines for those children we know received breastmilk as a child and those who know did not.
85. Download the csv data file from(assets/NISPUF17.csv) and write a program in python to see if there is any evidence of a link between vaccine effectiveness and sex of the child. Calculate the ratio of the number of children who contracted chickenpox but were vaccinated against it (at least one varicella dose) versus those who were vaccinated but did not contract chicken pox. Return results by sex.
86. Write a program in Python to read the file from metropolitan regions and associated sports teams from assets/wikipedia_data.html and calculate the win/loss ratio's correlation with the population of the city it is in for the NHL using 2018 data
87. Write a program in Python to read the file from metropolitan regions and associated sports teams from assets/wikipedia_data.html and calculate the win/loss ratio's correlation with the population of the city it is in for the NBA using 2018 data.
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89. Write a program in Python to read the file from metropolitan regions and associated sports teams from assets/wikipedia_data.html and calculate the win/loss ratio's correlation with the population of the city it is in for the NFL using 2018 data.
90. Write a program in Python to explore the hypothesis that given an area has two sports teams in different sports, those teams will perform the same within their respective sports.
91. Write a program in python to perform MapReduce to calculate a count of the number of squares, stars, circles, hearts and triangles in the data.